

BEAVER MINES, AB, CanadaWMO: **713040**Lat: **49.4678N**Lon: **114.1750W**Elev: **1257**StdP: **87.11**Time Zone: **-7.00 (NAM)**Period: **04-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-27.6	-24.3	-31.2	0.2	-27.2	-27.8	0.3	-23.9	13.1	-2.7	11.8	-1.4	1.3	170	0.636

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.1	28.8	18.1	27.0	16.9	25.3	16.0	22.3	26.2	20.6	24.1	18.7	22.0	3.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.0	18.3	25.2	19.2	16.3	22.7	17.2	14.4	20.9	72.1	25.7	65.6	24.3	58.2	22.2	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.1	9.5	8.3	DB	-33.0	32.1	5.2	2.3	-36.8	33.8	-39.9	35.1	-42.8	36.4	-46.6	38.1	(4)
(5)			WB	-32.5	19.8	5.3	4.2	-36.3	22.9	-39.4	25.4	-42.3	27.7	-46.2	30.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.0	-5.3	-8.0	-1.8	4.5	9.8	14.1	17.3	16.3	11.9	5.6	0.2	-5.4	(6)	
	DBStd	10.47	8.32	8.73	7.55	4.89	4.50	3.96	3.18	3.51	4.91	5.26	6.37	7.90	(7)	
	HDD10.0	2540	473	503	366	172	59	6	1	1	36	153	294	477	(8)	
	HDD18.3	4922	731	736	623	417	266	141	57	81	196	396	543	735	(9)	
	CDD10.0	725	1	0	1	6	54	128	227	198	93	17	2	0	(10)	
	CDD18.3	63	0	0	0	0	2	13	25	19	4	0	0	0	(11)	
	CDH23.3	880	0	0	0	1	28	139	381	284	43	4	0	0	(12)	
	CDH26.7	186	0	0	0	0	5	29	83	63	5	0	0	0	(13)	
(14) Wind	WSAvg	3.4	3.9	3.6	3.8	3.7	3.2	3.1	2.6	2.5	2.8	3.7	3.9	4.0	(14)	
(15) Precipitation	PrecAvg	583	38	30	35	56	77	82	46	54	60	31	37	37	(15)	
	PrecMax	840	98	97	69	143	181	293	165	144	175	82	88	77	(16)	
	PrecMin	329	7	6	2	8	19	9	4	2	3	1	0	5	(17)	
	PrecStd	137	22	19	17	29	41	64	42	37	43	24	21	22	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.2	10.2	14.1	22.1	26.8	29.5	30.7	31.0	27.1	23.0	16.1	9.8	(19)	
		MCWB	4.5	4.1	6.7	10.4	19.8	20.8	18.1	18.4	15.0	17.6	8.8	4.4	(20)	
	2%	DB	7.6	8.0	11.3	18.2	23.1	27.0	29.0	28.2	24.3	17.7	12.9	7.1	(21)	
		MCWB	3.2	3.4	4.6	8.6	12.8	18.8	17.9	16.3	15.0	10.5	7.9	2.6	(22)	
	5%	DB	5.9	6.3	9.3	14.9	20.9	24.7	27.1	26.5	22.2	15.4	10.0	5.3	(23)	
		MCWB	1.4	1.9	4.2	7.2	11.8	16.9	17.1	15.9	13.9	9.2	6.5	1.8	(24)	
	10%	DB	4.7	4.1	7.7	12.1	18.7	22.2	25.3	24.7	20.3	13.2	7.8	3.8	(25)	
		MCWB	0.6	0.1	2.9	5.7	10.9	15.5	16.2	14.9	13.2	7.9	4.5	0.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.0	4.7	9.0	11.5	19.7	23.5	24.0	24.1	20.0	17.8	12.6	6.0	(27)	
		MCDB	7.2	8.5	11.5	18.9	24.1	28.3	27.6	27.5	24.5	21.6	14.0	7.5	(28)	
	2%	WB	3.9	3.6	5.8	9.6	15.9	21.6	22.5	21.0	18.2	13.1	8.1	3.7	(29)	
		MCDB	7.2	7.6	10.1	15.3	20.9	25.2	26.2	25.0	21.2	15.7	11.8	6.4	(30)	
	5%	WB	2.2	2.2	4.2	7.9	13.2	19.1	21.0	19.0	16.7	10.7	6.6	1.8	(31)	
		MCDB	5.0	5.7	8.8	14.3	18.0	22.1	24.3	22.3	19.1	13.6	9.6	4.8	(32)	
	10%	WB	0.8	0.3	3.0	6.4	11.3	16.2	18.7	17.1	15.0	8.6	4.7	0.6	(33)	
		MCDB	4.1	3.8	7.4	11.6	17.3	21.0	21.9	21.0	17.5	12.7	7.6	3.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.4	8.6	8.9	9.7	10.6	10.6	13.1	13.1	10.6	8.5	7.6	7.3	(35)	
		MCDBR	7.4	6.7	10.2	16.2	15.5	14.1	16.3	16.3	14.8	11.9	10.7	6.9	(36)	
	5% WB	MCWBR	5.2	4.8	6.1	8.4	7.6	6.7	7.4	7.0	7.3	6.9	7.4	5.0	(37)	
		MCWBR	7.5	8.4	9.9	15.2	12.9	11.3	12.1	12.2	11.4	11.4	9.4	6.6	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.240	0.250	0.280	0.340	0.344	0.337	0.334	0.348	0.314	0.281	0.258	0.230	(40)	
		taud	2.391	2.375	2.393	2.269	2.372	2.434	2.456	2.412	2.487	2.562	2.501	2.414	(41)	
	Edn at Noon	Edn at Noon	844	918	940	905	914	921	918	888	891	874	818	809	(42)	
		Edh at Noon	62	81	96	122	115	109	105	105	88	68	56	52	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.14	2.12	3.23	4.59	5.48	5.98	6.87	5.58	3.91	2.36	1.23	0.90	(44)	
	RadStd	RadStd	0.11	0.17	0.29	0.35	0.46	0.47	0.48	0.39	0.38	0.25	0.11	0.08	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page